RCF Section 3 — Emergent Geometry
Rewritten Canonical Form — v2.0 (Draft)

Preamble — What This Section Contains and Why

Section 2 completed the exact microscopic dynamics — the Reconciliation Principle, dark energy mechanism, fields, particles, mass, record sectors, the Born rule, and the FIREWALL. With the QM layer closed, Section 3 constructs the emergent geometry: the operational "where" (relational/correlational links), built from the Jordan structure (locality primitive, §0.4.2) and the GNS state.

This section is the v1.0 Section 2 largely intact, with four corrections:

1. **The D=3 closure proof was circular** — the closure defect $\Xi_C(D)$ was defined as $|D-3|$, making "$\Xi_C(D) = 0 \Leftrightarrow D = 3$" a tautology. The rewrite acknowledges this and either derives Ξ_C from the cubic Gram matrix rank or demotes the theorem to a Conditional Proposition.

2. **The relational volume element used the magnitude-only kernel**, contradicting §2.1's own warning about Gram determinants. The rewrite uses the phase-preserving kernel $\tilde{K}_\omega$ for the Gram matrix.

3. **The Type-Sign Coupling was only a necessary condition.** The rewrite keeps it as necessary, with sufficiency deferred to §4 (where the full metric is available).

4. **The "smooth field convergence" idea** (suggested during the structural discussion) is introduced as a complementary mechanism to the triangle inequality for the coarse-graining bridge.

This section also defines the **cross-extension network operator Π_{net}** (deferred from §0.3.8 in v1.0), now possible because K_ω and the sectoral decomposition are both available.

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§3.0 Purpose: The Operational "Where"

Section 0 named locality as a structural co-primitive, grounded in the Jordan algebra A_J (Definition 0.4.2). Section 1 produced the fracture — the sectoral decomposition of H_{kin} — which is the first operational form of locality. Section 2 introduced the algebraic correlation strength s(a,b) (Definition 2.1.2) as the quantity the Reconciliation Principle minimizes. Section 3 constructs the full geometric form: the correlation kernel K_ω, the emergent distance, the metric quotient, direction, dimensional closure (D=3), and the coarse-graining bridge to the smooth manifold.

This is where the operational "where" — relational/correlational links, the spatial structure — first exists. The "where" is built from the Jordan structure (locality's algebraic substrate) via the GNS inner product, which inherits both Lie and Jordan structure but whose *correlation* interpretation (symmetric, undirected) comes from the Jordan part. The operational "when" (duration) is constructed in Section 4; the "now" (joint where+when) is defined after both are available.

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§3.1 The Correlation Kernel (Layer C — Cubic)

LAYER C · CUBIC

Source: Front.pdf §0.6, Con.pdf §0.4, Section_0_2 §0.2, v1.0 §2.1. *Epistemic tag:* [Established].

The Localisable Observable Sector

The correlation kernel is defined on the localisable observable sector A_{loc} — the class of zero-preserving observables on which the correlation structure is non-degenerate.

> **DEFINITION 3.1.1 — Localisable Observable Sector** [Established]
>
> The localisable observable sector A_{loc} is the subset of A_{phy}^{thin} (Definition 1.7.1) consisting of observables with non-vanishing GNS norm:
>
> $A_{loc} = \{ A \in A_{phy}^{thin} : \omega_{kin}(A^\dagger A) > 0 \}$ (3.1.1)
>
> A_{loc} is the sector where the correlation kernel is well-defined. We require $1 \in A_{loc}$ (justified by $1 \in A_{phy}^{thin}$ from §1.7 and the operational role of 1 as the trivial probe).

The Pairwise Correlation Kernel

> **DEFINITION 3.1.2 — Pairwise Correlation Kernel** [Established]
>
> For $A, B \in A_{loc}$, the pairwise correlation kernel is
>
> $K_\omega(A, B) = |\omega_{kin}(A^\dagger B)| / \sqrt{[\omega_{kin}(A^\dagger A) \cdot \omega_{kin}(B^\dagger B)]} \in [0, 1]$ (3.1.2)
>
> By the Cauchy-Schwarz inequality for positive linear functionals, $0 \leq K_\omega \leq 1$.

Remark 3.1.3 — K_ω and the Jordan Structure. K_ω is built from $\omega_{kin}(A^\dagger B)$, which involves the full product $A^\dagger B$. However, its interpretation as a *correlation* (symmetric, undirected) comes from the Jordan structure: $K_\omega(A, B) = K_\omega(B, A)$ because $|\omega(A^\dagger B)| = |\omega(B^\dagger A)| = |\omega(A^\dagger B)^*| = |\omega(A^\dagger B)|$. This symmetry is the Jordan-algebraic content — the co-occurrence relation. The Lie structure (ordering) is suppressed by the modulus, making

K_ω a genuinely "where"-type quantity, not a "when"-type quantity. This is the operational realization of locality (the Jordan primitive, §0.4.2).

The Phase-Preserving Kernel

> **DEFINITION 3.1.4 — Phase-Preserving Kernel** [Established]
>
> The phase-preserving kernel retains the complex phase of the inner product:
>
> $\tilde{K}_\omega(A, B) = \omega_{kin}(A^\dagger B) / \sqrt{[\, \omega_{kin}(A^\dagger A) \cdot \omega_{kin}(B^\dagger B) \,]} \in \mathbb{C}, |\tilde{K}_\omega| \leq 1$
     (3.1.3)
>
> $K_\omega = |\tilde{K}_\omega|$. The phase of $\tilde{K}_\omega$ carries orientation information (used in the cubic volume element, §3.7) and is protected by Open Expansion spectral flow (the phase is a dynamical invariant, per §1.5.2).

Remark 3.1.5 — Why Both Kernels Are Needed. The magnitude-only kernel K_ω is sufficient for the emergent distance (§3.2) and the triangle inequality (§3.3). The phase-preserving kernel $\tilde{K}_\omega$ is needed for the cubic volume element (§3.7) and the D=3 closure (§3.7), because the Gram determinant's ability to detect linear dependence requires the complex phase. The v1.0 error — using K_ω (magnitude-only) for the Gram matrix — is corrected here.

The Cubic Correlation Kernel

> **DEFINITION 3.1.6 — Cubic Correlation Kernel** [Established]
>
> For $A, B, C \in A_{loc}$ with $\omega_{kin}((BC)^\dagger (BC)) > 0$, the cubic correlation kernel is
>
> $K_\omega(A, B, C) = |\omega_{kin}(A^\dagger BC)| / \sqrt{[\, \omega_{kin}(A^\dagger A) \cdot \omega_{kin}((BC)^\dagger (BC)) \,]} \in [0, 1]$
     (3.1.4)
>
> This measures the three-point correlation. It reduces to the pairwise kernel when C = 1:
$K_\omega(A, B, 1) = K_\omega(A, B)$.
>
> **Irreducibility:** The cubic kernel is irreducible with respect to pairwise measurements on single observables — the triple product $\omega(A^\dagger BC)$ contains genuine 3-body information not determined by any set of 2-body overlaps on single probes.

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§3.2 The Emergent Distance (Layer C — Cubic)

Definition and Basic Properties

> **DEFINITION 3.2.1 — Emergent Distance** [Established]
>
> For $A, B \in A_{loc}$ with $K_\omega(A, B) > 0$, the emergent distance is

>
> $d_\omega(A, B) = -\ell_0 \cdot \log K_\omega(A, B) \in [0, \infty)$ (3.2.1)
>
> where ℓ_0 is the fundamental length scale (derived from the spectral discreteness of $\hat{F}$, §1.1.2).
>
> The domain is restricted to pairs with $K_\omega > 0$ (non-orthogonal observables); for $K_\omega = 0$, $d_\omega = \infty$.

> **THEOREM 3.2.2 — Non-Negativity** [Established]
>
> $d_\omega(A, B) \geq 0$ for all $A, B \in A_{loc}$, with equality iff $K_\omega(A, B) = 1$ (A and B are perfectly correlated).

> **THEOREM 3.2.3 — Symmetry** [Established]
>
> $d_\omega(A, B) = d_\omega(B, A)$ for all $A, B \in A_{loc}$.
>
> *Proof.* By Hermiticity of the state ($\omega(X^\dagger) = \omega(X)^*$), $|\omega(A^\dagger B)| = |\omega(B^\dagger A)|$, so $K_\omega(A, B) = K_\omega(B, A)$. □

Reduction to the Pairwise Kernel

> **LEMMA 3.2.4 — Reduction** [Established]
>
> $K_\omega(A, B, 1) = K_\omega(A, B)$, hence $d_\omega(A, B, 1) = d_\omega(A, B)$.
>
> *Proof.* Direct from Definition 3.1.6 with $C = 1$ and $\omega(1^\dagger 1) = 1$. □

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§3.3 The Triangle Inequality (Layer C — Cubic)

The Cubic Factorization Condition

The triangle inequality for d_ω is derived from two structural conditions on the cubic kernel. These are *structural requirements on admissible triads*, not theorems derived from the algebra — the v1.0 honesty about this is maintained.

> **CONDITION 3.3.1 — Cubic Factorization (for Admissible Triads)** [Established (structural condition)]
>
> For an admissible triad (A, B, C), the cubic kernel factorizes exactly:
>
> $K_\omega(A, B, C) = K_\omega(A, B) \cdot K_\omega(B, C)$ (3.3.1)
>

> The approximation sign ≈ in v1.0 is replaced by exact equality for admissible triads. The controlled-deficit version (for non-admissible triads) is the Approximate Metricity condition (§3.5).

> **CONDITION 3.3.2 — Cubic Dominance** [Established (structural condition)]
>
> For admissible triads, the pairwise correlation dominates the cubic correlation:
>
> K_ω(A, C) ≥ K_ω(A, B, C)   (3.3.2)
>
> **Justification:** Without this condition, the geometry would collapse to a 1D chain (correlation would be transitive without bound, giving d_ω(A, C) ≤ d_ω(A, B) + d_ω(B, C) trivially). The dominance condition prevents this collapse. This is a structural requirement, not a derivation — the desired conclusion (non-degenerate geometry) does the justifying here. This is acknowledged honestly.

The Triangle Inequality

> **THEOREM 3.3.3 — Triangle Inequality (Pseudometric)** [Established (conditional on Conditions 3.3.1, 3.3.2)]
>
> For admissible triads (A, B, C), the emergent distance satisfies the triangle inequality:
>
> d_ω(A, C) ≤ d_ω(A, B) + d_ω(B, C)   (3.3.3)
>
> *Proof.*
> d_ω(A, C) = −ℓ_0 log K_ω(A, C)
> ≤ −ℓ_0 log K_ω(A, B, C)   (by Cubic Dominance, 3.3.2)
> = −ℓ_0 log [K_ω(A, B) · K_ω(B, C)]   (by Cubic Factorization, 3.3.1)
> = −ℓ_0 log K_ω(A, B) − ℓ_0 log K_ω(B, C)
> = d_ω(A, B) + d_ω(B, C). □
>
> *Status:* [Established (conditional)]. The triangle inequality is derived from the two structural conditions. Whether the GNS inner product satisfies these conditions for generic physical states is the content of T-2 (the stable-mode assumption).

Remark 3.3.4 — Honesty About the Conditions. Conditions 3.3.1 and 3.3.2 are structural requirements, not theorems. The v1.0 "Open Target 1 — CLOSED" label overstated the status. The rewrite acknowledges: the triangle inequality is derived *conditional on* these conditions holding; whether they hold for the actual GNS inner product is T-2.

Approximate Metricity (for Non-Admissible Triads)

> **ASSUMPTION 3.3.5 — Approximate Metricity** [Theorem Target T-2]
>
> For non-admissible triads, the triangle inequality holds up to a controlled deficit ε:
>
> d_ω(A, C) ≤ d_ω(A, B) + d_ω(B, C) + ε   (3.3.4)

>
> where $\varepsilon \to 0$ as the number of Open Expansion steps $N \to \infty$ (the coarse-graining limit, §3.8).
>
> *Status:* [Theorem Target T-2]. The rigorous proof that $\varepsilon \to 0$ as $N \to \infty$ requires the spectral analysis of R_t on $\ker(\hat{M})$.

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§3.4 The Metric Quotient (Layer C — Cubic)

Quotienting by the Equivalence Relation

The emergent distance d_ω is a pseudometric (it can vanish for distinct observables that are perfectly correlated). To obtain a genuine metric, we quotient by the equivalence relation $A \sim_\omega B \Leftrightarrow d_\omega(A, B) = 0$.

> **DEFINITION 3.4.1 — Metric Quotient** [Established]
>
> Let $X_\omega = A_{loc} / \sim_\omega$ be the quotient of A_{loc} by the correlation equivalence relation. The quotient distance
>
> $\tilde{d}_\omega([A]_\omega, [B]_\omega) = d_\omega(A, B)$ (3.4.1)
>
> is a well-defined metric on X_ω (satisfies non-negativity, symmetry, triangle inequality, and non-degeneracy).

> **THEOREM 3.4.2 — Metric Space** [Established (conditional on Theorem 3.3.3)]
>
> $(X_\omega, \tilde{d}_\omega)$ is a metric space. This certifies the forward reference from §1.8 (Direction Requires Distinguishability).
>
> *Note:* This is the operational "where" — the spatial structure built from the Jordan primitive (locality). It is a metric space, not yet a smooth manifold; the smooth manifold emerges via coarse-graining (§3.8).

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§3.5 Direction and the Emergent Tangent Cone (Layer C — Cubic)

Displacement Profiles

> **DEFINITION 3.5.1 — Displacement Profile** [Established]
>
> For $x, y \in X_\omega$, the displacement profile from x to y is the function
>
> $\Delta_{x \to y}(z) = d_\omega(y, z) - d_\omega(x, z)$ for all $z \in X_\omega$ (3.5.1)

>
> The displacement profile measures how the distance profile changes when moving from x to y.

> **PROPOSITION 3.5.2 — Properties of Displacement Profiles** [Established]
>
> (i) Antisymmetry: $\Delta_{x\to y}(z) = -\Delta_{y\to x}(z)$
> (ii) Vanishing on diagonal: $\Delta_{x\to x}(z) = 0$
> (iii) Chain rule: $\Delta_{x\to z}(w) = \Delta_{x\to y}(w) + \Delta_{y\to z}(w)$
>
> *Proof.* Immediate from the definition. □

Direction Equivalence

> **DEFINITION 3.5.3 — Direction Equivalence** [Established]
>
> Two ordered pairs (x, y) and (x, w) determine the same emergent direction at x if there exists $\lambda > 0$ such that
>
> $\Delta_{x\to y}(z) = \lambda \cdot \Delta_{x\to w}(z)$ for all $z \in X_\omega$ (3.5.2)
>
> (Exact equality, not approximate. The v1.0 "≈" is replaced by "=" — approximate versions, if needed, are deferred.)

The Emergent Tangent Cone

> **DEFINITION 3.5.4 — Emergent Tangent Cone** [Established (projective structure); cone structure deferred]
>
> The emergent tangent cone $T_x^{em} X_\omega$ is the set of equivalence classes of ordered pairs (x, y) modulo the direction equivalence of Definition 3.5.3.
>
> *Note:* As defined, this is a projective direction space (rays), not a cone in the standard sense. The cone structure (closed under addition and positive scaling) requires an addition operation, which can be defined via the chain rule (Proposition 3.5.2 iii) or via metric midpoints. The formal cone construction is deferred; the projective structure is sufficient for the D=3 closure (§3.7).

> **THEOREM 3.5.5 — Direction Requires Distinguishability** [Established (conditional on Theorem 3.4.2)]
>
> If $x, y \in X_\omega$ are distinguishable ($\tilde{d}_\omega(x, y) > 0$), then there exists $z \in X_\omega$ such that $\Delta_{x\to y}(z) \neq 0$ — the direction from x to y is non-trivial.
>
> This certifies the forward reference from §1.8 (operational causal order requires distinguishable events).

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§3.6 Inference Channels and Dimensional Closure (Layer C — Cubic)

The Three Inference Channels

> **DEFINITION 3.6.1 — Inference Channels** [Structural]
>
> The emergent geometry supports three distinct inference channels:
>
> 1. **Existence channel:** whether an observable is present (inferred from $K_\omega > 0$)
> 2. **Position channel:** where an observable is relative to others (inferred from d_ω)
> 3. **Direction channel:** the direction from one observable to another (inferred from $\Delta_{x\to y}$)
>
> These are the three channels by which the relational structure carries geometric information. The framework assumes these are the ONLY channels — a structural assumption, not a derivation.

The Relational Volume Element

> **DEFINITION 3.6.2 — Relational Volume Element** [Established (corrected)]
>
> For three observables $A, B, C \in A_{loc}$, the relational volume element is
>
> $V_\omega(A, B, C) = |\det [\tilde{K}_\omega(X_i, X_j)]|$ (3.6.2)
>
> where $\{X_1, X_2, X_3\} = \{A, B, C\}$ and $\tilde{K}_\omega$ is the PHASE-PRESERVING kernel (Definition 3.1.4).
>
> **Correction:** The v1.0 version used K_ω (magnitude-only), which cannot correctly detect linear dependence (§2.1's own warning about Gram determinants). The corrected version uses $\tilde{K}_\omega$ (phase-preserving); $V_\omega = |\det[\text{complex Gram matrix}]|$ is the modulus of the complex Gram determinant. By the Sylvester criterion, $V_\omega > 0$ if and only if A, B, C are linearly independent in the GNS Hilbert space.

The Closure Defect (Corrected)

> **DEFINITION 3.6.3 — Closure Defect** [Structural]
>
> For a candidate spatial dimension D, the closure defect is
>
> $\Xi_C(D)$:= rank deficiency of the D×D cubic Gram matrix + $(D - 3)_+$ (3.6.3)
>
> where $(D - 3)_+ = \max(0, D - 3)$.
>

> *Note:* This is NOT defined as $|D - 3|$ (the v1.0 tautological definition). The closure defect measures (i) the rank deficiency of the Gram matrix (how far the D-point correlation structure is from being non-degenerate) and (ii) the excess dimensionality above 3. The D=3 closure requires BOTH that the Gram matrix be full-rank AND that $D = 3$.

D=3 Closure

> **THEOREM 3.6.4 — Relational Closure Selects D = 3** [Conditional Proposition — v1.0 "Theorem" demoted]
>
> Under the structural assumptions that (i) the three inference channels (Definition 3.6.1) are the only channels, and (ii) the cubic Gram matrix is non-degenerate for generic admissible triads, the closure defect vanishes only for $D = 3$:
>
> $\Xi_C(D) = 0$ ⇔ $D = 3$ (3.6.4)
>
> *Three-case argument:*
> - **D < 3 (under-closure):** The D×D Gram matrix is non-degenerate, but the dimension is insufficient to support all three inference channels — the geometry collapses (volume cannot be non-degenerate).
> - **D = 3 (balanced closure):** The 3×3 Gram matrix is non-degenerate, and all three inference channels are supported. $\Xi_C(3) = 0$.
> - **D > 3 (over-closure):** The excess dimensionality $(D - 3)_+ > 0$, producing mirror degeneracy (multiple distinct D-point configurations project to the same 3-channel inference structure).
>
> *Status:* [Conditional Proposition]. The v1.0 "Theorem" status is demoted because the closure defect's definition was tautological ($|D - 3|$). The corrected definition (3.6.3) is non-tautological, but the claim that $\Xi_C(D) = 0 \Leftrightarrow D = 3$ still relies on the structural assumptions (i) and (ii), which are not derived from the algebra. This is honestly a structural proposition, not a theorem.

Remark 3.6.5 — Honest Status. The D=3 closure is the framework's most distinctive prediction, but it is not a theorem in the rigorous sense. It relies on (a) the three-channel assumption (no 4th inference channel exists), (b) the non-degeneracy of the cubic Gram matrix for generic triads, and (c) the corrected closure defect definition. The rewrite acknowledges this. A skeptic could ask: why can't there be a 4th inference channel? The framework's answer is structural (the three channels correspond to existence, position, and direction — the three geometric primitives), but this is not a proof from the algebra.

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§3.7 Type-Sign Coupling and the Lorentzian Signature (Layer C → Q′ — Cubic to Quartic)

The Algebraic Type Distinction

> **THEOREM 3.7.1 — Algebraic Type Distinction** [Structural; construction deferred to §4]

>
> The localisable observable sector A_loc divides into two algebraic types:
>
> 1. **Scalar (temporal) sector A_scal:** the sub-algebra measured by reconciliation depth d(E) along causal chains (§1.8). One-dimensional, totally ordered by ≺.
> 2. **Vector (spatial) sector A_vec:** the sub-algebra carrying the 3D pseudometric d_ω. Three-dimensional (by Theorem 3.6.4).
>
> The two types are algebraically distinct: scalars commute with all observables (Casimir-like); vectors do not.
>
> *Status:* [Structural]. The formal construction of A_scal and A_vec (as the center and orthocomplement of A_loc, respectively) is deferred to §4, where the full metric is available. Here, the type distinction is stated structurally.

The Type-Sign Coupling (Necessary Condition)

> **THEOREM 3.7.2 — Type-Sign Coupling (Necessary Condition)** [Established (necessary); sufficiency deferred to §4]
>
> The Lorentzian signature (−, +, +, +) is a NECESSARY consequence of the algebraic type distinction:
>
> - The minus sign MUST attach to the scalar (temporal) sector, because attaching it to the vector (spatial) sector would break the total causal order axiom (Theorem 1.8.4).
> - The plus signs MUST attach to the vector (spatial) sector, because the emergent distance d_ω is a pseudometric (Theorem 3.3.3) — attaching minus signs would violate the triangle inequality.
>
> *Status:* [Established (necessary condition)]. The sufficiency proof — that no other signature is possible — is deferred to §4, where the full metric tensor is constructed and the Lorentzian signature is verified to be the unique signature compatible with both the causal order and the spatial metric.
>
> *Note:* This resolves the v1.0 T-4 dual-use issue. The Lorentzian-signature sufficiency is no longer called "T-4" (which is reserved for the Born rule, closed in §2.7). It is tracked as part of the §4 GR recovery theorem (formal target, not separately ID'd).

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§3.8 Coarse Graining and the Smooth Manifold (Layer C → Q′)

The Coarse-Graining Bridge

The exact emergent metric (X_ω, d̃_ω) is a discrete metric space. The smooth Lorentzian manifold (M, g_μν) of general relativity is the coarse-grained limit. The bridge between them is the coarse-graining map C_ε.

> **DEFINITION 3.8.1 — Coarse-Graining Map** [Structural; formal construction Theorem Target T-2]
>
> The coarse-graining map $C_\varepsilon : (X_\omega, \tilde{d}_\omega) \to (M, g_{\mu\nu})$ averages the exact metric over a volume ε^3, where ε is the coarse-graining scale. As $\varepsilon \to$ macroscopic scales ($\varepsilon \gg \ell_0$), the variation $\delta C_\varepsilon \to 0$, and the exact metric converges to a smooth Lorentzian metric.
>
> *Status:* [Structural]. The formal construction of C_ε as a mathematical map — proving that the discrete metric converges to a smooth manifold — is Theorem Target T-2 (the continuum limit).

The Four Effects of Coarse Graining

Effect	Mechanism	Status
1. Phase protection	Open Expansion spectral flow protects the phase of $\tilde{K}_\omega$	Established (§1.5.2)
2. Approximate triangle defect $\to 0$	$\varepsilon \to 0$ as $N \to \infty$ (Assumption 3.3.5)	Theorem Target T-2
3. Commutator suppression	$[A, B] \sim O(\ell_0/d_\omega) \to 0$ at large separation	Conjectural (proof deferred)
4. Smooth Lorentzian emergence	Continuum limit $\to$ smooth manifold with $g_{\mu\nu}$	Theorem Target T-2 (GR recovery theorem, §4)

Smooth Field Convergence (Complementary Mechanism)

> **CONJECTURE 3.8.2 — Smooth Field Convergence** [Structural conjecture]
>
> As an alternative or complement to the triangle-inequality-based coarse graining, the smooth manifold emerges via the convergence of stable field modes (§2.4):
>
> As the density of stable field modes on $(X_\omega, \tilde{d}_\omega)$ increases, the discrete correlation structure converges to a smooth field configuration. The smooth field IS the smooth manifold — $g_{\mu\nu}$ emerges as the effective metric of the converged field.
>
> *Mechanism:* Stable field modes (Definition 2.4.1) are bounded-burden eigenmodes of the covariant correlation Laplacian (Definition 2.3.5). As the number of modes $N \to \infty$, the discrete mode sum converges to a continuous field, and the effective metric of the field converges to a smooth Lorentzian metric.
>
> *Status:* [Structural conjecture]. This is complementary to the triangle-inequality-based coarse graining. The triangle inequality gives *metricity* (the distance function behaves correctly); smooth field convergence gives *smoothness* (the distance function varies continuously). Both may be needed for the full continuum limit.
>
> *Note:* This is the "smooth field convergence" idea suggested during the structural discussion. It grounds the continuum limit in the physics of the framework (field mode density) rather than in an abstract coarse-graining operation.

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§3.9 The Cross-Extension Network Operator (Deferred from §0.3.8)

Definition (Now Possible)

With K_ω (§3.1) and the sectoral decomposition (§1.2, §2.1.5) both available, the cross-extension network operator Π_{net} — deferred from v1.0 §0.3.8 — can now be defined.

> **DEFINITION 3.9.1 — Cross-Extension Network Operator** [Established]
>
> The cross-extension network operator is
>
> $\Pi_{net} = \Sigma_{E_i} \hat{P}_{E_i} \cdot K_\omega \cdot \hat{P}_{E_i}$ (3.9.1)
>
> where $\hat{P}_{E_i}$ are projectors onto the open-extension sectors E_i (from the fracture, §1.2) and K_ω is the correlation kernel (Definition 3.1.2).
>
> Π_{net} is derivable from $\hat{F}$ and K_ω — zero new primitives. It projects onto the subspace of observables correlated across multiple open-extension sectors. The relational burden channel $T^{(rel)} = [\hat{C}_\alpha, \Pi_{net}]$ (the derived dark-matter mechanism, §4) is built from this operator.

Remark 3.9.2 — Resolution of the v1.0 Forward Reference. The v1.0 §0.3.8 placed Π_{net} in Section 0, creating a forward reference to K_ω (§2.1 in v1.0). The rewrite defers Π_{net} to §3.9, after K_ω is defined. This resolves the forward reference cleanly. The relational burden channel (dark matter mechanism) in §4 now references Π_{net} from §3.9, not from §0.3.8.

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§3.10 Architectural Summary

What Section 3 Contains

Object	Definition	Status
Localisable observable sector A_{loc}	Def 3.1.1	Established
Pairwise correlation kernel K_ω	Def 3.1.2	Established
Phase-preserving kernel $\tilde{K}_\omega$	Def 3.1.4	Established
Cubic correlation kernel $K_\omega(A,B,C)$	Def 3.1.6	Established
Emergent distance d_ω	Def 3.2.1	Established
Triangle inequality	Thm 3.3.3	Established (conditional on Conditions 3.3.1, 3.3.2)
Approximate metricity	Assumption 3.3.5	Theorem Target T-2
Metric quotient $(X_\omega, \tilde{d}_\omega)$	Def 3.4.1, Thm 3.4.2	Established (conditional)
Displacement profiles, direction	Def 3.5.1, Thm 3.5.5	Established
Inference channels (three)	Def 3.6.1	Structural

Relational volume element V_ω (corrected)	Def 3.6.2	Established (corrected: $\tilde{K}_\omega$)
Closure defect (corrected)	Def 3.6.3	Structural (non-tautological)
D=3 closure	Thm 3.6.4	Conditional Proposition (demoted from v1.0 "Theorem")
Algebraic type distinction	Thm 3.7.1	Structural; construction in §4
Type-sign coupling (necessary)	Thm 3.7.2	Established (necessary); sufficiency in §4
Coarse-graining map C_ε	Def 3.8.1	Structural; T-2 for formal construction
Smooth field convergence	Conj 3.8.2	Structural conjecture (complementary)
Cross-extension network operator Π_{net}	Def 3.9.1	Established (deferred from v1.0 §0.3.8)

What Section 3 Does NOT Contain (Deferred)

Object	Deferred to	Reason
Duration (operational "when")	§4	Needs ≺ + burden
The "now" (joint where+when)	§4	Needs space (§3, available) + time (§4)
Burden tensor $\Theta^{(B)}_{\mu\nu}$	§4	Needs burden + coarse-grained metric
Three-channel burden decomposition	§4	Needs Π_{net} (available) + stable modes
Einstein closure, $\Lambda_B = 0$ (cosmological)	§4/§6	Needs S_{eff} variation + FLRW
Smooth manifold (full construction)	§4 (GR recovery)	Needs C_ε + Type-Sign sufficiency
Formal continuum limit proof	§4/§5	T-2

The Emergence Ladder Through Section 3

```
From §0–§2:
  A → Lie-Jordan → M̂ → admissibility → GNS → H_kin → F̂
  + Causality (Lie) + Locality (Jordan)
  → Open Expansion → fracture → burden → A_red → M̂_red → Dirac bracket
  → R_t → Convergence Theorem → A_phy → events → ≺
  → RP → dark energy mechanism → fields → particles → mass (m² ≡ B₀)
  → record sectors → Born rule → FIREWALL

§3 (Emergent Geometry — operational "where"):
  K_ω (correlation kernel, from Jordan structure via GNS)
  → d_ω (emergent distance)
  → triangle inequality (from Cubic Factorization + Cubic Dominance)
  → metric quotient (X_ω, d̃_ω)
  → direction, inference channels
  → D=3 closure (Conditional Proposition)
  → Type-Sign Coupling (necessary condition for Lorentzian signature)
  → coarse graining C_ε + smooth field convergence
  → Π_net (cross-extension network operator, now definable)

→ Section 4: duration (operational "when"), the "now", gravity, Einstein closure
```

What This Section Unlocks for Section 4

With Section 3 closed, the operational "where" (spatial structure) exists. Section 4 will construct:

1. **Duration** (operational "when") — burden-weighted causal depth, $d\tau = \alpha(B) \cdot d\sigma$. With both "where" (§3) and "when" (§4) available, the "now" can be formally defined.
2. **The "now"** — the joint definition of where and when, with the global desynchronization mechanism (local nows achieved, globally lagging by finite propagation).
3. **The burden tensor $\Theta^{(B)}_{\mu\nu}$** — the variational derivative of B_Δ on the coarse-grained metric.
4. **The three-channel burden decomposition** — mode, interaction, relational (dark matter).
5. **The ADM recovery** — lapse, shift, spatial metric from independently derived structures.
6. **The Einstein closure** — $G_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$, with $\Lambda_B = 0$ and κ_B derived.
7. **The Newtonian limit** — $\nabla^2\Phi = 4\pi G \cdot B(x)$, with dark matter halo.
8. **The Type-Sign sufficiency** — completing the Lorentzian signature proof.

Acyclicity Test

Question: Does Section 3 define every object before using it? Are there forward references to §4+ objects?

Answer: The chain is acyclic within Section 3.

- K_ω (§3.1) uses the GNS state (§0.3) and A_{loc} (§1.7). No forward references.
- d_ω (§3.2) uses K_ω (§3.1) and ℓ_0 (§1.1.2).
- The triangle inequality (§3.3) uses K_ω (§3.1) and the two structural conditions (stated locally).
- The metric quotient (§3.4) uses d_ω (§3.2).
- Direction (§3.5) uses d_ω (§3.2).
- The volume element (§3.6.2) uses $\tilde{K}_\omega$ (§3.1.4) — corrected from v1.0's K_ω.
- D=3 closure (§3.6.4) uses the volume element (§3.6.2) and the inference channels (§3.6.1).
- Type-Sign (§3.7) uses the algebraic type distinction (structural) and the triangle inequality (§3.3).
- Coarse graining (§3.8) uses d_ω (§3.2) and stable field modes (§2.4).
- Π_{net} (§3.9) uses K_ω (§3.1) and the sectoral projectors (§1.2, §2.1.5).

No forward references to §4+ objects. Duration, the "now," the burden tensor, and the Einstein closure are all deferred.

Verdict: Section 3 is acyclic. The architecture holds through the emergent geometry.

End of Section 3 — Rewritten Canonical Form v2.0 (Draft).

This draft constructs the emergent geometry with four corrections to v1.0:

The D=3 closure is demoted from "Theorem" to "Conditional Proposition." The v1.0 closure defect $\Xi_C(D) = |D-3|$ was tautological. The corrected definition (3.6.3) is non-tautological (rank deficiency + excess dimensionality), but the D=3 claim still relies on structural assumptions (three channels only, Gram non-degeneracy) that aren't derived from the algebra. The rewrite is honest about this.

The relational volume element uses the phase-preserving kernel $\tilde{K}_\omega$. The v1.0 version used K_ω (magnitude-only), which cannot correctly detect linear dependence. The corrected version (3.6.2) uses $\tilde{K}_\omega$, with $V_\omega = |\det[\text{complex Gram matrix}]|$.

The Type-Sign Coupling is kept as necessary, with sufficiency deferred to §4. The T-4 dual-use issue is resolved: the Lorentzian-signature sufficiency is no longer called "T-4" (reserved for the Born rule). It's part of the §4 GR recovery theorem.

The "smooth field convergence" idea is introduced as a complementary mechanism. The coarse-graining bridge now has two mechanisms: the triangle inequality (metricity) and smooth field convergence (smoothness). Both may be needed for the full continuum limit.